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About

Dr. Jinwei Liu received his Ph.D. from the Department of Computer Science and Engineering, The Chinese University of Hong Kong, under the supervision of Prof. Evangeline F.Y. Young. Previously, he got his bachelor's degree in Computer Science and Technology from Sichuan University. His research interests include Electronic Design Automation (EDA), Combinatorial Optimization, Machine Learning and Reinforcement Learning. He has won multiple awards in world-renowned EDA contests, including four championships at ICCAD contests and one championship at an ISPD contest. He also received the best paper award at ISPD 2023.

Research Interests

- Electronic Design Automation
- Combinatorial Optimization
- Machine Learning
- Reinforcement Learning

Selected Publications

- Jinwei Liu, Evangeline F.Y. Young "EDGE: Efficient DAG-based Global Routing Engine" In 2023 60th ACM/IEEE Design Automation Conference (DAC).
- Fangzhou Wang*, Jinwei Liu*, Evangeline F.Y. Young "FastPass: Fast Pin Access Analysis with Incremental SAT Solving." In 2023 ACM International Symposium on Physical Design (ISPD).
- Jinwei Liu, Xiaopeng Zhang, Shiju Lin, Xinshi Zang, Jingsong Chen, Bentian Jiang, Martin D.F. Wong, Evangeline F.Y. Young "Partition and place finite element model on wafer-scale engine." In 2022 59th ACM/IEEE Design Automation Conference (DAC).
- Shiju Lin, Jinwei Liu, Tianji Liu, Martin D.F. Wong, Evangeline F.Y. Young "NovelRewrite: node-level parallel AIG rewriting." In 2022 59th ACM/IEEE Design Automation Conference (DAC).
- Shiju Lin, Jinwei Liu, Martin D.F. Wong. “GAMER: GPU Accelerated Maze Routing.” In 2021 IEEE/ACM International Conference On Computer Aided Design (ICCAD).
- Jinwei Liu, Gengjie Chen, Evangeline F.Y. Young. REST: Constructing Rectilinear Steiner Minimum Tree via Reinforcement Learning.” In 2021 58th ACM/IEEE Design Automation Conference (DAC).
- Bentian Jiang, Jingsong Chen, Jinwei Liu, Lixin Liu, Fangzhou Wang, Xiaopeng Zhang, and Evangeline F.Y. Young. “CU. POKer: Placing DNNs on Wafer-Scale Al Accelerator with Optimal Kernel Sizing.” In 2020 IEEE/ACM International Conference On Computer Aided Design (ICCAD).

- Jinwei Liu, Chak-Wa Pui, Fangzhou Wang, and Evangeline F.Y. Young. “CUGR: Detailed-Routability-Driven 3D Global Routing with Probabilistic Resource Model.” In 2020 57th ACM/IEEE Design Automation Conference (DAC).